# Kushagra Jain

Pune, Maharashtra, India | +91-6388943930 | kushagrajainaj2003@gmail.com
linkedin.com/in/kushagra-jain-957300259 | github.com/Kushagra2418

**AI Engineer | Data Scientist**

## SUMMARY

Versatile and proactive Full-Stack Web Developer and AI/ML Engineer with hands-on experience building scalable web applications, GenAI microservices, and production-grade RAG and multi-agent systems. Skilled in developing RESTful APIs, securing authentication, and delivering intuitive front-end solutions, alongside expertise in LLM orchestration, vector search, and AI deployment. Adept at problem-solving in C++ and Python, with a strong commitment to writing clean, maintainable code.

## EDUCATION

**D.Y. Patil College of Engineering, Akurdi, Pune** 2022 – 2026
B.Tech in Computer Science and Engineering; Cumulative GPA: 8.2

## PROJECTS

**RAG-Based Technical Documentation Assistant** Jul 2026
*LangGraph (StateGraph), FastAPI, Groq LLM, Sentence-Transformers | GitHub:* RAG-Based-Technical-Documentation-Assistant

- Built a self-corrective Retrieval-Augmented Generation (RAG) system that answers questions over a corpus of technical documentation, architected as a LangGraph StateGraph with dedicated nodes for query analysis, retrieval, document grading, and answer generation, and served through a FastAPI application
- Designed a self-correction loop that grades every retrieved chunk for relevance and rewrites the query to retry retrieval (bounded by a retry limit) when results are poor, falling back to an honest "I don't know" instead of hallucinating an answer
- Ran LLM inference on Groq with embeddings computed locally via sentence-transformers, so the only external credential required is a free Groq API key; added optional Self-RAG-style hallucination verification and a web-search fallback

**Agentic RAG Ecommerce Electronic Shop** May 2026
*LangGraph.js, Groq (Llama 3.3 70B), Hybrid RAG (Vector + SQL) | GitHub:* Agentic-Rag-Ecommerce

- Built an AI-powered local electronics marketplace combining hybrid Retrieval-Augmented Generation (vector + SQL) with a LangGraph.js agent running on Groq's Llama 3.3 70B
- Delivered intelligent product discovery, multi-seller price comparison, and location-aware shopping, all from a single conversational interface

**AI Fashion Outfit Recommendation System** Mar 2026
*Python, FAISS, FashionCLIP, Gemini, Streamlit — DareXAI ML & AI Engineer Intern Assignment | GitHub:* AI-Fashion-Outfit-Recommendation-System

- Built a conversational outfit recommender that parses free-text style requests and an optional user profile, then assembles complete, gender-consistent outfits slot-by-slot from a 68-item catalogue using FAISS vector retrieval
- Designed a custom, explainable compatibility-scoring engine blending image and text similarity (FashionCLIP), pairing priors learned from 25 stylist-curated outfits, colour harmony, and occasion match — keeping the LLM scoped to intent parsing and explanation generation
- Engineered graceful degradation across CLIP/offline embedding backends and Gemini/rule-based intent parsing, so the full pipeline runs end-to-end with no GPU, internet access, or API key required

## TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, SQL
- **GenAI & LLM:** RAG (Basic, Multi-Agent, Hierarchical), LLM Fine-tuning (SFT, DPO, LoRA), Prompt Engineering
- **AI Systems:** LangChain, LangGraph, AI Agents, Multi-Agent Systems, A2A Protocols, LLM Orchestration
- **Data Science & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, OpenCV
- **Machine Learning:** Supervised Learning (Regression, Classification, Decision Trees, Random Forest, SVM), Unsupervised Learning (Clustering, Dimensionality Reduction, PCA), Deep Learning (CNNs, RNNs/LSTMs, Transformers, PyTorch, TensorFlow)
- **Databases:** SQL
- **Other Tools:** Git, GitHub, Postman, JWT, Bcrypt, Docker, GCP, AWS (Basics), Hugging Face, Streamlit, FastAPI

## CERTIFICATIONS

- MERN Stack Development — 100xDevs (Dec 2024)
- Fundamentals of Deep Learning — NVIDIA (Mar 2025)

## LANGUAGES & INTERESTS

- **Languages:** Hindi, English
- **Interests:** Swimming, Cricket, Badminton